



Republic of the Philippines
Isabela State University
Echague, Isabela



College of Computing Studies, Information and Communication Technology

WEEK 2:

Development Environment Setup and Initial Coding

Dupitas, John Mitchel M.
Iddurut, Justin R.
Ignacio, Dave Raphael M.



ACTIVITY 1 DEVELOPMENT ENVIRONMENT CONFIGURATION

FORM 1 DEVELOPMENT ENVIRONMENT CHECKLIST

Directions

Check all software successfully installed and configured.

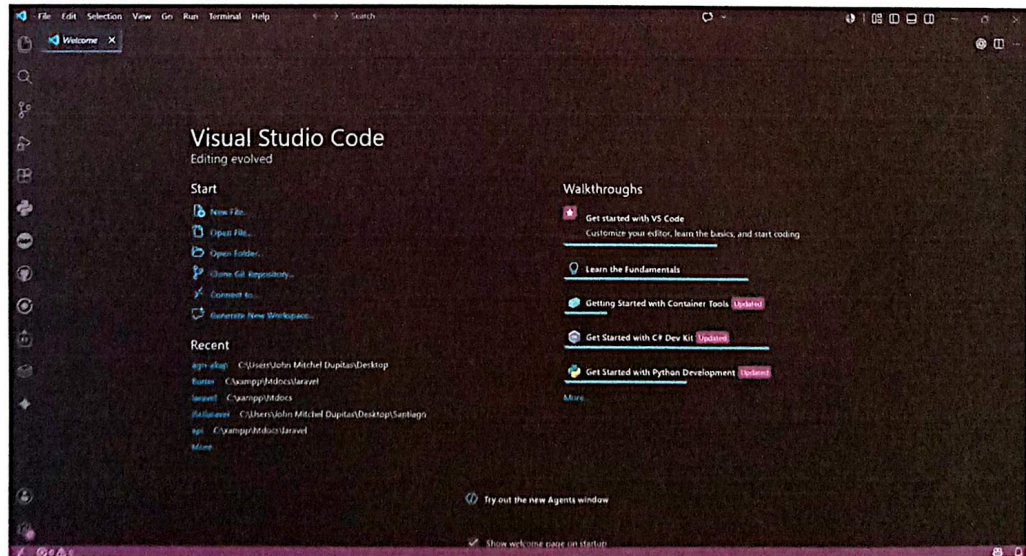
Tool/Software	Required	Installed	Version
IDE	☒	☒	Visual Studio Code v1.89+
Programming Language Runtime	☒	☒	PHP v8.4+ , Node.js v23.x (LTS) / TypeScript v5.x
Database Server	☒	☒	MariaDB v10+
Version Control Tool	☒	☒	Git v2.47+
Framework	☒	☒	Laravel v11.x (Backend) , Ionic Vue v7.x + Capacitor v5.x (Frontend)
API Testing Tool	☒	☒	Postman v10.x

Environment Test Results

Component Tested	Result
IDE Launch	Pass
Database Connection	Pass
Repository Access	Pass
Framework Execution	Pass

Required Evidence

- ☐ Screenshot of Installed Tools
 - ☐ Visual Studio Code







□ Screenshot of Running Development Environment

```
PS C:\Users\John Mitchel Dupitas\Desktop\agri-akap\agri-akap-frontend> ionic serve
> vite.cmd --host=localhost --port=8100
[vite] ✓ Console Ninja extension is connected to Vite, see https://tinyurl.com/2vt8jxzw
[vite] → Local: http://localhost:8100/
[vite] → press h + enter to show help

[INFO] Development server running!

Local: http://localhost:8100

Use Ctrl+C to quit this process

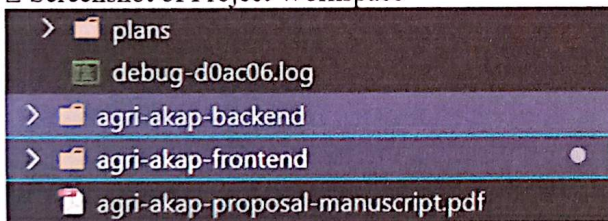
PS C:\Users\John Mitchel Dupitas\Desktop\agri-akap\agri-akap-backend> php artisan serve

[INFO] Server running on [http://127.0.0.1:8000].

Press Ctrl+C to stop the server

2026-07-28 21:24:29 /api/dashboard/stats ..... ~ 1s
2026-07-28 21:24:57 /api/login ..... ~ 1s
2026-07-28 21:24:58 /api/login ..... ~ 5s
2026-07-28 21:26:25 /api/farmers/barangays ..... ~ 505.89ms
2026-07-28 21:26:26 /api/farmers/barangays ..... ~ 1s
2026-07-28 21:26:33 /api/programs ..... ~ 1s
2026-07-28 21:26:34 /api/farmers/barangays ..... ~ 1s
2026-07-28 21:26:34 /api/programs ..... ~ 2s
```

□ Screenshot of Project Workspace





ACTIVITY 2
DATABASE IMPLEMENTATION

FORM 2
DATABASE IMPLEMENTATION REPORT

Directions

Document the database created for the project.

Project Database Name: db_agri_akap

Database Platform: MariaDB v10.11 (Relational DBMS)

Core Database Tables

Table Name	Purpose
users	Stores accounts for MAO Admins and Field Technicians with Sanctum auth token management.
farmers	Centralized RSBSA-compliant registry storing demographic data, QR codes, and PWD/Senior status.
programs	Manages agricultural assistance campaigns (e.g., RCEF seeds, fertilizer allocations).
distributions	Records subsidy releases; includes composite unique keys to mathematically prevent double-claiming.
farm_plots	Tracks geotagged farm plot boundaries, land area, and crop types.
crop_monitorings	Logs crop standing stages (vegetative to harvest) and yield observations.
pest_outbreaks	Captures field scouting reports for pest and disease outbreaks.

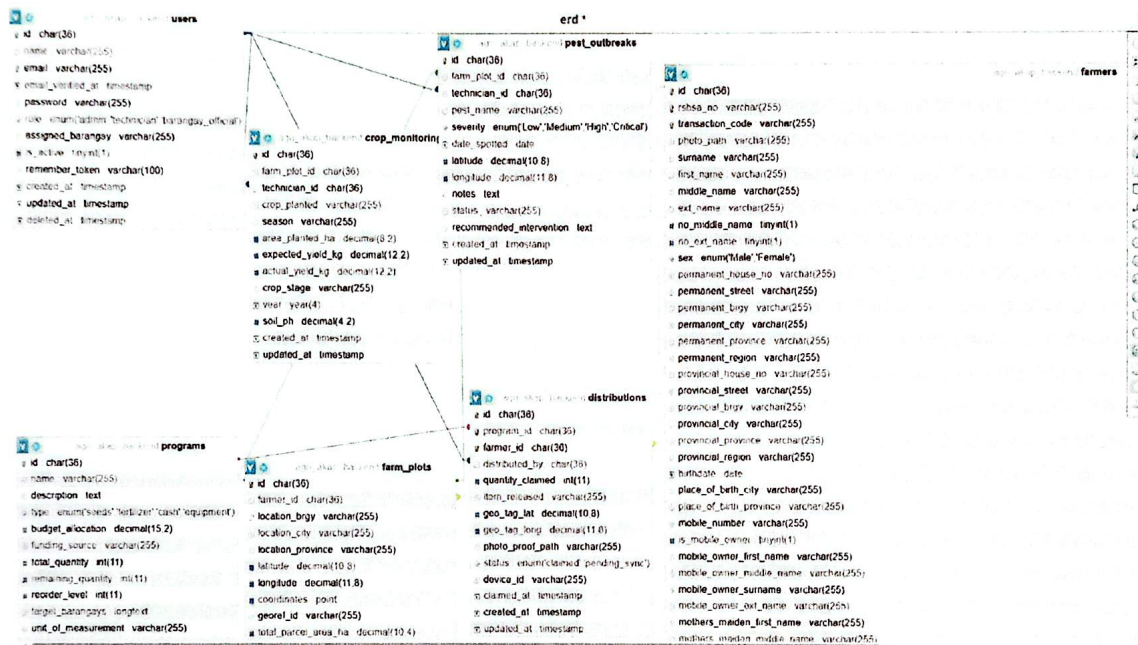
Relationships Implemented

Relationship	Status
One-to-One	Implemented
One-to-Many	Implemented
Many-to-Many	Implemented



Required Evidence

□ ERD Diagram



□ Database Screenshot

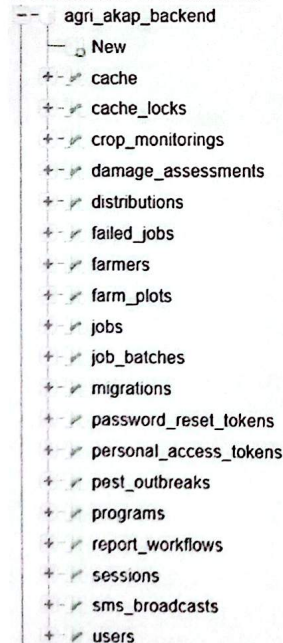




Table Structure Screenshot

Table	Action	Rows	Type	Collation	Size	Overhead
cache	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 K	
cache_locks	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 K	
crop_monitorings	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 K	
damage_assessments	Browse Structure Search Insert Empty Drop	14	InnoDB	utf8mb4_unicode_ci	112.0 K	
distributions	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	60.0 K	
failed_jobs	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	60.0 K	
farmers	Browse Structure Search Insert Empty Drop	15	InnoDB	utf8mb4_unicode_ci	24.0 K	
farm_plots	Browse Structure Search Insert Empty Drop	20	InnoDB	utf8mb4_unicode_ci	32.0 K	
jobs	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 K	
job_batches	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 K	
migrations	Browse Structure Search Insert Empty Drop	28	InnoDB	utf8mb4_unicode_ci	16.0 K	
password_reset_tokens	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 K	
personal_access_tokens	Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_unicode_ci	64.0 K	
pest_outbreaks	Browse Structure Search Insert Empty Drop	14	InnoDB	utf8mb4_unicode_ci	30.0 K	
programs	Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_unicode_ci	16.0 K	
report_workflows	Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_unicode_ci	30.0 K	
sessions	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 K	
sms_broadcasts	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	16.0 K	
users	Browse Structure Search Insert Empty Drop	68	InnoDB	utf8mb4_unicode_ci	48.0 K	
19 tables	Sum	257	InnoDB	utf8mb4_general_ci	816.0 K	0 B

Connectivity Test Screenshot

```
PS C:\Users\John Mitchell Dupitas\Desktop\agri-akap> cd agri-akap-backend
PS C:\Users\John Mitchell Dupitas\Desktop\agri-akap\agri-akap-backend> php artisan db:monitor

Database name ..... Connections [1] OK
mariadb

PS C:\Users\John Mitchell Dupitas\Desktop\agri-akap\agri-akap-backend> php artisan migrate:status

Migration name ..... Batch / Status
0001_01_01_000009 create_users_table ..... [1] Ran
0001_01_01_000001 create_cache_table ..... [1] Ran
0001_01_01_000002 create_jobs_table ..... [1] Ran
2026_05_27_231824 programs ..... [1] Ran
2026_05_27_231943 farmers ..... [1] Ran
2026_05_27_232033 distributions ..... [1] Ran
2026_05_27_232032 farm_plots ..... [1] Ran
2026_06_06_015207 create_personal_access_tokens_table ..... [1] Ran
2026_06_09_233842 create_sms_broadcasts_table ..... [1] Ran
2026_07_01_232826 create_crop_monitorings_table ..... [1] Ran
2026_07_01_232910 create_pest_outbreaks_table ..... [1] Ran
2026_07_02_000113 create_damage_assessments_table ..... [1] Ran
2026_07_11_000001 add_haraway_official_role_to_users_table ..... [1] Ran
```

ACTIVITY 3 FRAMEWORK AND PROJECT STRUCTURE SETUP

FORM 3 PROJECT STRUCTURE REPORT

Directions

Document the project architecture.

Folder	Purpose
frontend	Houses the unified Ionic Vue 3 + Capacitor source code for Web Admin and Mobile Android app.
backend	Holds Laravel REST API code, Sanctum middleware, business logic controllers, and routes.
database	Contains Laravel database migrations, seeders, and MariaDB relational schema definitions.
documentation	Stores project documentation, API contracts, ERD diagrams, and technical specifications.
testing	Contains PHPUnit backend API tests and Cypress/Jest E2E frontend test suites.



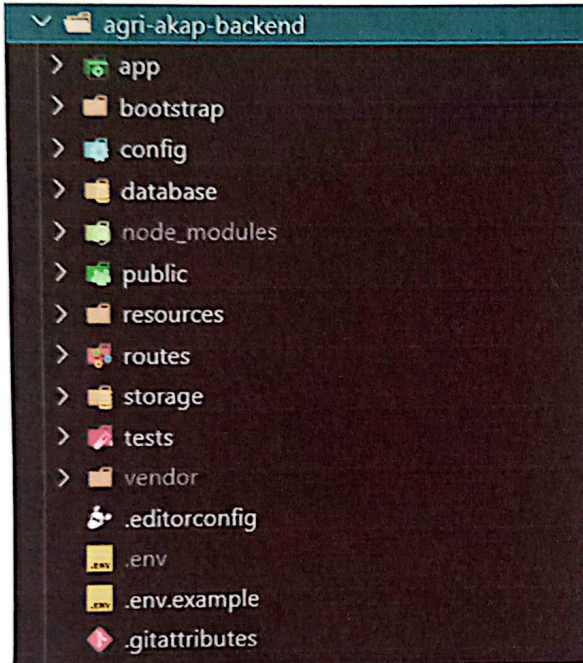
Framework Configuration

Framework & Version: ☐ Backend Framework: Laravel v11.x (PHP 8.2+)

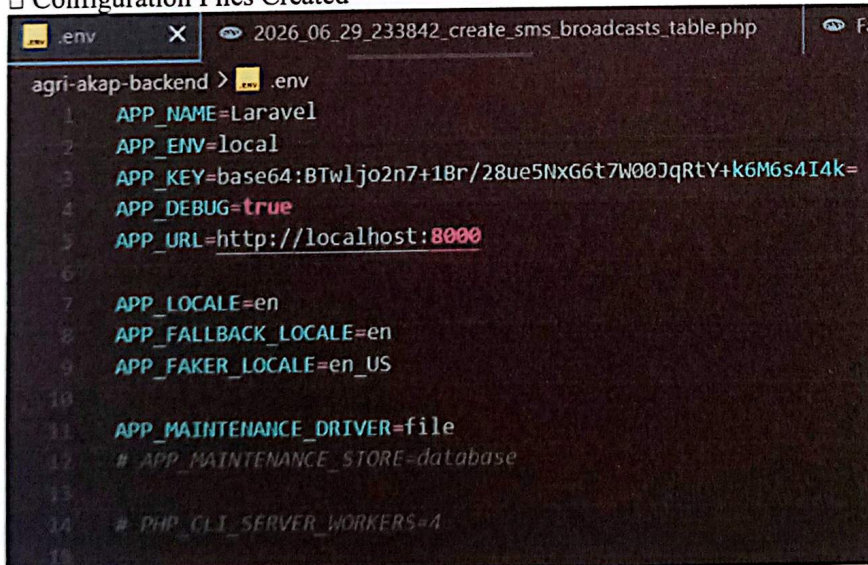
☐ Frontend Framework: Ionic v7.x with Vue 3 & Capacitor v5.x

Project Structure Validation

☐ Folder Structure Created

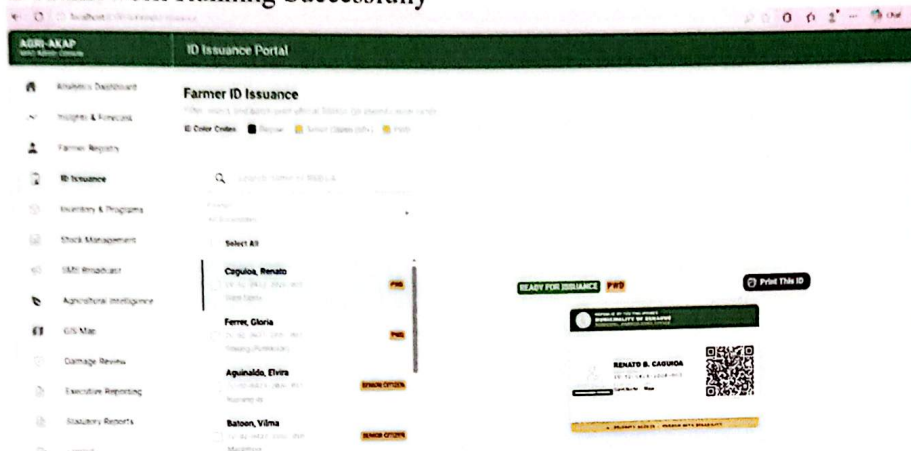


☐ Configuration Files Created





□ Framework Running Successfully



ACTIVITY 4
FOUNDATIONAL MODULE DEVELOPMENT

FORM 4
INITIAL MODULE DEVELOPMENT REPORT

Directions

Document all foundational modules completed.

Module	Description	Status
User Authentication	Laravel Sanctum API login token issuance paired with Vue Router role guards.	Complete
Database Connection Module	Eloquent ORM mapping to MariaDB schema with UUID auto-generation.	Complete
Dashboard Layout & Navigation	Responsive layout supporting Web Admin Sidebar and Mobile Navigation Bar.	Complete
QR Farmer Identification	Capacitor Camera integration scanning QR codes to pull farmer profile records.	Complete

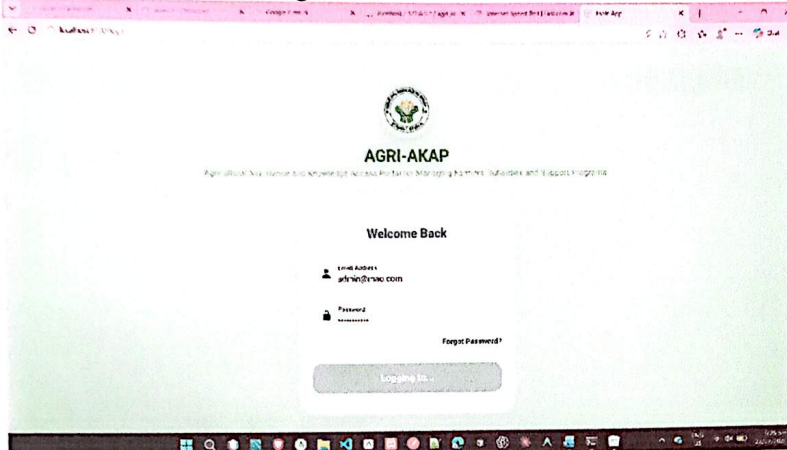
Module Testing

Test Performed	Result
Login Function Test	Pass
Database Connection Test	Pass
Navigation Test	Pass

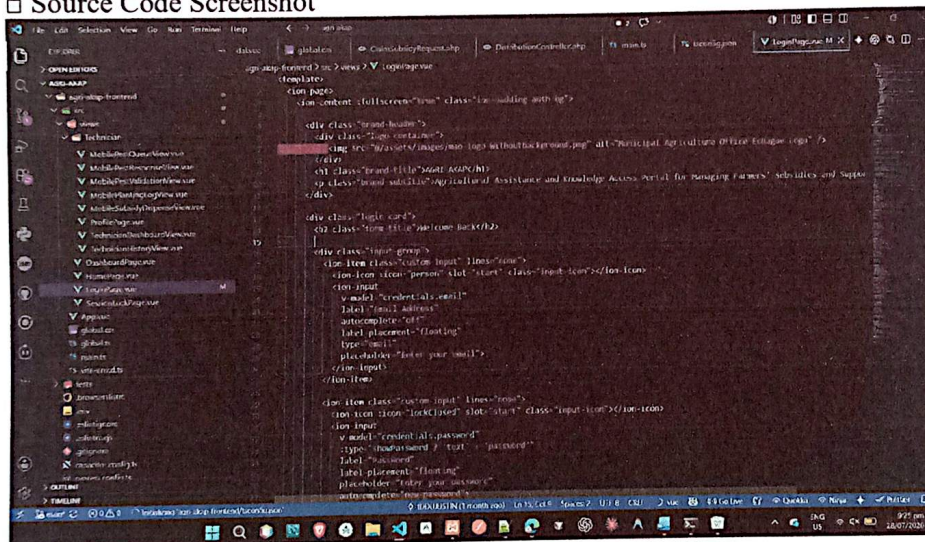


Required Evidence

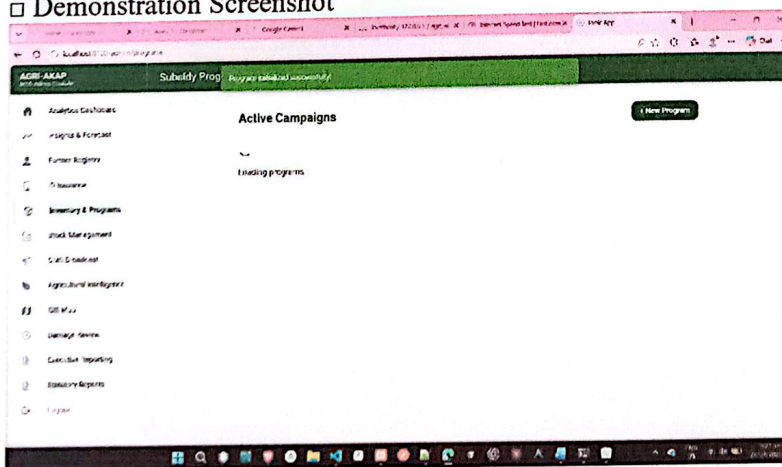
□ Screenshots of Running Module



□ Source Code Screenshot



□ Demonstration Screenshot





ACTIVITY 5
CODING STANDARDS ESTABLISHMENT

FORM 5
CODING STANDARDS AGREEMENT

Naming Conventions

Item	Standard
Variables	camelCase
Functions	camelCase
Classes	PascalCase
Tables	snake_case (Plural)
Files	PascalCase (Vue) / snake_case (Laravel/PHP)

Documentation Standards

- ☐ Source Code Comments Required: PHPDoc block comments on controller methods; JSDoc on Vue Pinia store functions.
- ☐ Function Documentation Required: Detailed parameters, return types, and potential API error status codes.
- ☐ Database Documentation Required: Migration file descriptions and ERD data dictionary maintained.

Repository Standards

Commit Format: Conventional Commits (type(scope): message)

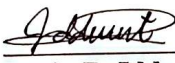
Branching Strategy: GitFlow / Feature-Branch Workflow

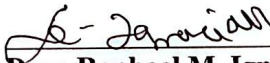
Team Agreement

All members agree to follow the coding standards established above.

Member Signatures:


John Mitchel M. Dupitas


Justin R. Iddurut


Dave Raphael M. Ignacio



ACTIVITY 6 REPOSITORY INITIALIZATION

FORM 6 INITIAL SOURCE CODE REPOSITORY REPORT

Repository Name: agri-akap-frontend | agri-akap-backend

Repository URL:

<https://github.com/JohnMitchelDupitas/agri-akap-frontend.git>

<https://github.com/JohnMitchelDupitas/agri-akap-backend.git>

Team Access

Member	Access Granted
John Mitchel M. Dupitas	Yes
Justin R. Iddurut	Yes
Dave Raphael M. Ignacio	Yes

Initial Commits

Commit Description	Date
Initial commit(Frontend)	Apr 20, 2026
Initial commit(Backend)	May 26, 2026

Required Evidence

☐ Repository Homepage Screenshot

agri-akap-frontend Public

Pin Watch 0

main 4 Branches 0 Tags Go to file Add file Code

JohnMitchelDupitas added mobile features pest and disaster 3a59d51 · last week 21 Commits

vscode	Registration Form	2 months ago
public	Initial commit	3 months ago
src	added mobile features pest and disaster	last week
tests	Initial commit	3 months ago
browserdistrc	Initial commit	3 months ago
env	site manifest high	last month
eslintignore	Initial commit	3 months ago
eslinttrc.js	added sidebar & vils	last month
gitignore	Initial commit	1 months ago
capacitor.config.ts	first build	2 weeks ago
cypress.config.ts	Initial commit	3 months ago
index.html	Initial commit	3 months ago
ionic.config.json	Initial commit	3 months ago



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Initial Commit Screenshot

Commits on Apr 20, 2020

Initial commit

John Mitchel Dupitas comm Med on Apr 20

bd46527

Project Structure Screenshot

JohnMitchelDupitas removed pdc and added more reports needed			32e4841 - yesterday	13 Commits
app	removed pdc and added more reports needed			yesterday
bootstrap	firstdraft8			last week
config	fix bugs			last week
database	removed pdc and added more reports needed			yesterday
public	initial commit			2 months ago
resources	initial commit			2 months ago
routes	removed pdc and added more reports needed			yesterday
storage	initial commit			2 months ago
tests	initial commit			2 months ago
.editorconfig	initial commit			2 months ago
.env.example	firstdraft8			last week
.gitattributes	initial commit			2 months ago
.gitignore	initial commit			2 months ago
.npmrc	initial commit			2 months ago
README.md	initial commit			2 months ago
artisan	initial commit			2 months ago
composer.json	add auth			last month
composer.lock	add auth			last month
package-lock.json	add auth			last month
package.json	add auth			last month
phpunit.xml	initial commit			2 months ago
vite.config.js	initial commit			2 months ago

Team Access Screenshot

Select all

Type Find a collaborator...

☐	 dabe dabierphil10 • Collaborator	
☐	 Iddurut, Justin R. IDDUJUSTIN • Collaborator	
☐	 John Mitchel Dupitas johnmitcheldupitas • Collaborator	



ACTIVITY 7
ADVISER VALIDATION

Purpose

To verify readiness for full-scale system development.

Procedure

1. Present all Week 2 outputs.
2. Demonstrate development environment.
3. Demonstrate database implementation.
4. Demonstrate initial modules.
5. Obtain adviser approval.

FORM 7
ADVISER VALIDATION RECORD

Item Evaluated	Status
Development Environment	☐ Approved ☒ For Revision
Database Implementation	☒ Approved ☐ For Revision
Initial Modules	☒ Approved ☐ For Revision
Coding Standards	☒ Approved ☐ For Revision
Repository Setup	☒ Approved ☐ For Revision

Adviser Recommendations

Review all system/ modules label. Use appropriate words.

Action Items

Adviser Signature: _____

Date: 2/11/26



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Review all system/ modules label. Use appropriate words.

Action Items

Adviser Signature: _____

Date: 8/16/26